

Hexadecimal · IG · Practice

U04 · 24 written + 6 pen-first = 30 marks · Name: _____

All questions are authored. Show working where requested. Respect the specified width; no calculator or converter is needed.

1. Ours: change the representation

(a) Convert binary **10110110** to hexadecimal. [1]

(b) Write hexadecimal **07** as an 8-bit binary value. [1]

(c) Convert denary **230** to hexadecimal. Show your working. [2]

(d) Convert hexadecimal **D08** to denary. Show your working. [2]

2. Ours: why this base?

(a) Give one benefit to a person of using hex to display a binary value. [1]

(b) Explain why one hexadecimal digit can represent exactly four bits. [1]

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3. Ours: colour bytes · exam-shape rehearsal

An RGB colour uses **#4EA209**. Give the 8-bit binary value for each byte. Our scheme awards one mark per correct nibble.

(a) Hexadecimal **4E** in 8-bit binary: [2]

(b) Hexadecimal **A2** in 8-bit binary: [2]

(c) Hexadecimal **09** in 8-bit binary: [2]

4. Ours: two other uses

Colour codes and MAC addresses have already been given. State two **other** uses of hexadecimal. [2]

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5. Ours: preserve the value and the width

(a) A 16-bit register holds **0101 1010 0011 1110**. Write its four hexadecimal digits.

[2]

(b) Convert denary **351** to hexadecimal. Show your working.

[2]

(c) Convert binary **10101101** to denary. Show your working.

[2]

(d) A student writes binary **11** for hex **03**. State whether the numerical value is correct. Then give the binary answer required for an **8-bit register**.

[2]

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6. Ours: build a translator · pen first [6]

 **Design before code.** Translate a binary string into hex by grouping bits and using the nibble table. No built-in base-conversion functions. These six marks are human marked on paper.

(a) Draw a flowchart or write numbered steps for binary-to-hex conversion. Include what happens when the bit count is not a multiple of four. [2]

(b) Trace your method for binary **110111001**. Show the padded groups and the hex output. [2]

(c) Translate hex **0F** into binary, preserving four bits per supplied hex digit. Explain why the first four bits are zero. [2]

 **Optional coding challenge · 0 marks.** Implement both directions in **challenge_U04_HexTranslator.py**. Pen first; code when ready. The file supplies the contract and local checks. No `int()`, `hex()` or `bin()` in your converter functions.

Above and beyond · unmarked. In RGB colour **#2864B4**, find the three denary channel values. Increase only the blue channel by 16 (denary); write the new colour code. This is an 8-bit-per-channel RGB representation, not every colour format.

Additional space - label the question number.
